Constant regret with finite preparation in Markov games

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Abstract

We study the computational formulation of catalogue problem C73-113 under its rational input convention and observable-action feedback. Players reconstruct their local tables, communicate through observed actions, and compute a rational coarse equilibrium over pure Markov plans. A maximum-deficit schedule tracks its weights with uniformly bounded error. This gives constant cumulative external regret against all history-dependent within-episode policies, with polynomial work per episode. The finite preparation period is included in the regret bound and may be very large as a function of the game.

1 The stated target

There are n players, a finite state set S, nonempty finite action sets A_i, a horizon $H \geq 1$, and a fixed initial state s_1. Rewards $r_{i,h}(s,a)$ lie in $[0,1]$, and $P_h(\cdot \mid s,a)$ is the next-state distribution. We use the catalogue's computational convention: the finite game data are rational and binary encoded [1, Common conventions, Computational representations]. Let L be their description length. The labeled finite structure is public, while reward and transition entries are initially accessible through the prescribed feedback only.

At a queried (h,s,a_i), player i receives its own reward averaged against the opponents' current Markov policies and the corresponding averaged transition distribution, both exactly. All realized states and actions are observed. A behavioral policy may depend on the entire observed history within an episode. Let Π_i be this policy class, and let $V_i(\pi)$ be expected total reward in one episode. Thus $0 \leq V_i(\pi) \leq H$. The target is

$$R_i(T) = \sup_{\tau_i \in \Pi_i} \sum_{t=1}^{T} (V_i(\tau_i, \pi_{-i}^t) - V_i(\pi^t)) \leq C_G \qquad (T \geq 1),$$

where τ_i is one fixed within-episode policy and C_G is finite and independent of T. Neither C_G nor total preparation time is required to be polynomial in L. The resource requirement is polynomial work and polynomially many feedback queries in each episode, as a function of the game description and the current episode number.

The source motivates constant regret and supplies the own-feedback oracles [2, §§2.3–2.5 and 5]. The catalogue specifies the ordinary external-regret target above. The construction below uses a finite initial calculation and action signals, and its bound is stated explicitly in terms of that calculation.

Theorem 1. *One deterministic decentralized procedure, independent of the final number of episodes, achieves the target for every game in this model. Every played profile is pure Markov. Each player makes at most one feedback query per episode, and its bit operations are bounded by a fixed polynomial in L and the current episode number. More precisely, the procedure has a finite preparation duration T_0 and computes a distribution with finite support size K, for which*

$$R_i(T) \leq HT_0 + 2H(K-1) \qquad \textit{for every } i \textit{ and } T \geq 1.$$

If every action set is a singleton, play the unique profile forever; all regrets are zero. Otherwise, call players with $|A_i| \geq 2$ active. A remaining player's regret is always zero, so its reward table is irrelevant to the computation below.

2 A rational distribution covering every deviation

The played plans depend only on (h, s); comparison policies may also depend on earlier states and actions. Let Σ_i be all pure Markov plans $[H] \times S \to A_i$, and put $\Sigma = \prod_i \Sigma_i$. Let D_i be all pure policies on the finite set of player i's pre-action histories, including histories of zero probability. Histories contain past states and joint actions and the current state, before the current actions are selected. Define

$$g_{i,d}(\sigma) = V_i(d, \sigma_{-i}) - V_i(\sigma) \qquad (d \in D_i,\ \sigma \in \Sigma).$$

These are rational numbers computable by finite enumeration of play histories. Consider the finite linear system

$$\mu_\sigma \geq 0, \qquad \sum_{\sigma \in \Sigma} \mu_\sigma = 1, \qquad \sum_{\sigma \in \Sigma} \mu_\sigma g_{i,d}(\sigma) \leq 0 \quad (i \text{ active},\ d \in D_i). \tag{1}$$

This system is nonempty. To prove it, set $v_{i,H+1}(s) = 0$ and work backward. At (h, s) choose a mixed Nash equilibrium of the finite stage game with payoffs

$$r_{i,h}(s, a) + \sum_{s'} P_h(s' \mid s, a) v_{i,h+1}(s'),$$

and let $v_{i,h}(s)$ be its expected payoff. Such equilibria exist by Nash's finite-game theorem [3]. The resulting behavioral Markov profile x resists every history-dependent unilateral deviation. Indeed, at any history ending in (h, s), the opponents' current actions have the prescribed stage distribution. By induction, a deviator's future payoff is at most $v_{i,h+1}$ at the next state. The stage Nash inequality then bounds its current reward plus continuation by $v_{i,h}(s)$. This proves the assertion backward through all histories.

Independently sample an action from $x_{i,h}(s)$ for each player and each (h, s), thereby obtaining a random member of Σ. Its law implements x, including against any fixed history-dependent deviation: a given stage-state pair is visited at most once, and its unused action sample is independent of the samples that determined the past. The induced distribution on Σ satisfies (1). It is used only to prove nonemptiness.

The feasible set is a nonempty compact rational polytope, so it has a rational vertex. For completeness, successively maximizing each coordinate on the preceding maximizing face produces a vertex. At a vertex, active constraint normals span the whole coordinate space; otherwise a small perturbation in both directions orthogonal to them would preserve all constraints and contradict extremality. A full-rank active rational system therefore determines that vertex by Gaussian elimination. This also gives a finite exact algorithm: enumerate all full-rank active systems, solve them, test all constraints, and keep the first passing solution in a fixed public enumeration order.

Discard zero weights and write its support as $\sigma^1, \ldots, \sigma^K$, with positive rational weights $\mu_1, \ldots, \mu_K$. For any behavioral τ_i, independently pre-sampling its action at each finite history gives a distribution on D_i that implements τ_i against every fixed σ_{-i}. The distribution is the same for all σ. Averaging the corresponding rows of (1) proves

$$\sum_{j=1}^{K} \mu_j (V_i(\tau_i, \sigma^j_{-i}) - V_i(\sigma^j)) \leq 0 \qquad (\tau_i \in \Pi_i). \tag{2}$$

Singleton-action players satisfy this with equality. Thus the supported profiles are Markov, while the comparison class retains all histories.

3 Legal decentralized preparation

Fix public orders on players, states, actions and table entries, and one baseline action for each player. Enumerate $(h, s, a) \in [H] \times S \times \prod_i A_i$, reserving one episode for each triple. In that episode every player j uses the constant Markov plan a_j. Each active player i queries (h, s, a_i) once. The opponents' policy is a point mass, so the reply is exactly $r_{i,h}(s, a)$ and $P_h(\cdot \mid s, a)$. After these finitely many episodes, each active player has its own reward table and the same full transition table. All queries concern the actual current opponent policy, including when the queried state is unreachable in that episode.

Active players next broadcast their reward tables, in public player order. For a sender i, fix two distinct actions labeled zero and one. It sends a bit through its action at $(1, s_1)$; every other action follows the baseline. Everyone observes the bit with certainty, independently of subsequent state transitions. These signaling plans are pure Markov. No player queries another player's reward oracle.

Use canonical finite binary encodings of rational entries and the fixed table order. A payload of length $b \geq 1$ is prefixed by $1^\ell 0$ $\mathrm{bin}(b)$, where $\ell = \lfloor \log_2 b \rfloor + 1$ and $\mathrm{bin}(b)$ has ℓ bits. Receivers count the initial ones, read the length and then the payload. All players therefore know exactly when the next sender starts. All these strings have finite length, polynomial in L under the stated finite-table encoding. Preparing, sending and reading them has polynomial cost per episode. A sole active player already has every relevant reward table; the same protocol is harmless in that case.

Every active player now has identical input for (1). Each runs the same deterministic exact program just described, executing one transition of a fixed binary Turing machine per episode while playing the baseline plan. This finite program eventually halts, in the same episode for every active player. Its simulation includes generating the linear system, solving it and writing the support plans and weights. Let T_0 count all exploration, signaling and computation episodes, including input and output preparation. It is finite and depends only on the game and the fixed procedure.

4 A schedule with uniformly bounded error

For $m \geq 0$, let $N_j(m)$ count uses of σ^j in the first m episodes after preparation, and put

$$d_j(m) = m\mu_j - N_j(m).$$

Start with all counts zero. In episode $m \geq 1$, choose the least index maximizing $d_j(m-1) + \mu_j$, play σ^j, and increment its count. All active players have the same weights, counts and tie rule, so they select the same joint plan and execute their own components.

For every m, the deficits sum to zero and each is greater than -1. The latter holds initially. Before the next selection, the numbers $d_j(m-1) + \mu_j$ sum to one, so the selected number is positive. Subtracting one leaves it greater than -1; unselected numbers have only increased. Induction proves the claim. If $K = 1$ the deficit is zero. Otherwise at most $K - 1$ coordinates are negative, and each has magnitude less than one. Zero total sum consequently gives

$$\sum_j |d_j(m)| \leq 2(K-1). \tag{3}$$

Fix any player i and any $\tau_i \in \Pi_i$, and abbreviate $g_j = V_i(\tau_i, \sigma^j_{-i}) - V_i(\sigma^j)$. We have $|g_j| \le H$ and $\sum_j \mu_j g_j \le 0$ by (2). Its total deviation gain after preparation is

$$\sum_j N_j(m) g_j = m \sum_j \mu_j g_j - \sum_j d_j(m) g_j \le H \sum_j |d_j(m)| \le 2H(K-1).$$

Each preparation episode contributes at most H. This proves the bound in Theorem 1, also when $T \le T_0$. The bound is uniform in τ_i, so taking its supremum proves the claim. By linearity, the ordinary uniform empirical mixture has coarse-deviation gain at most $\{HT_0 + 2H(K-1)\}/T$.

5 Per-episode resources

Exploration uses one own-feedback query per active player per episode; later phases use none. Communication and its counters have polynomial bit cost. During the simulated exact computation, at most $O(\mathrm{poly}(L) + t)$ tape cells have been initialized by episode t. Even direct tape scans therefore implement each simulated transition with polynomially many bit operations in L, t.

The final support and weights were written before episode T_0. Their total encoded length, and hence K, are bounded by the same tape bound. Write $\mu_j = p_j/q_j$ with $q_j > 0$. The next comparison value is

$$d_j(m-1) + \mu_j = \frac{mp_j - N_j(m-1)q_j}{q_j}.$$

Counts use $O(\log m)$ bits. Exact cross multiplication, scanning the stored support and retrieving the selected local Markov plan all have polynomial cost in its stored length and $\log m$, hence in L, t. This establishes a single per-episode polynomial bound for the entire procedure. All preparation is charged to T_0; the resulting $C_G = HT_0 + 2H(K-1)$ is finite but may grow rapidly with the game size. The algorithm never uses the final episode count T.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] Economics problem C73-113, Constant regret in general-sum Markov games. Statement and Common conventions, Computational representations. Problem statement.

[2] A. E. Yorulmaz and T. Başar. Near Optimal Convergence to Coarse Correlated Equilibrium in General-Sum Markov Games. 2025. arXiv:2511.02157v1, §§2.3–2.5 and 5.

[3] J. F. Nash, Jr. Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences* 36(1) (1950), 48–49. doi:10.1073/pnas.36.1.48.